

2010 – 2025 MARKET INTELLIGENCE

Global Electric Vehicle Market Analysis

Analyzing EV sales, adoption, powertrains, and charging infrastructure worldwide

Python

Pandas

Jupyter Notebook

Power BI

What Are We Trying to Understand?

- 1 How has global EV sales changed from 2010 to 2025?
- 2 Which countries are driving EV adoption?
- 3 How are BEVs and PHEVs performing against each other?
- 4 How has EV stock evolved over time?
- 5 Is charging infrastructure keeping pace with EV adoption?

OBJECTIVE

Transform raw EV data into actionable insights about market growth, adoption, and regional performance.

DATASET & SCOPE

From Raw Data to Analysis-Ready Data

RAW DATASET

49,375

rows × 9 columns

region_country	category	parameter
mode	powertrain	year
unit	value	Aggregate group

Includes historical data plus projection categories extending to 2035.

ANALYSIS FOCUS

- EV Sales
- EV Stock
- EV Sales Share
- EV Charging Points

Passenger-car analysis then narrowed to BEV and PHEV powertrains.

How Was the Data Prepared?



15 duplicate rows removed; records with missing region_country dropped.

22,605 records retained for the selected EV parameters.

PYTHON LIBRARIES

Pandas — Cleaning & transformation

NumPy — Data operations

Matplotlib — Visualization

Turning Data into Analysis



KEY ANALYSES PERFORMED

- | | | |
|-------------------------|---------------------------|---------------------|
| ✓ Global EV sales trend | ✓ Top 3 EV Market | ✓ BEV vs PHEV sales |
| ✓ Global EV stock | ✓ Charging infrastructure | ✓ EV market share |
| ✓ Top EV markets | ✓ China vs India vs USA | ✓ YOY GROWTH |

EV Adoption Has Shifted into High Gear

7,400

EV sales in 2010

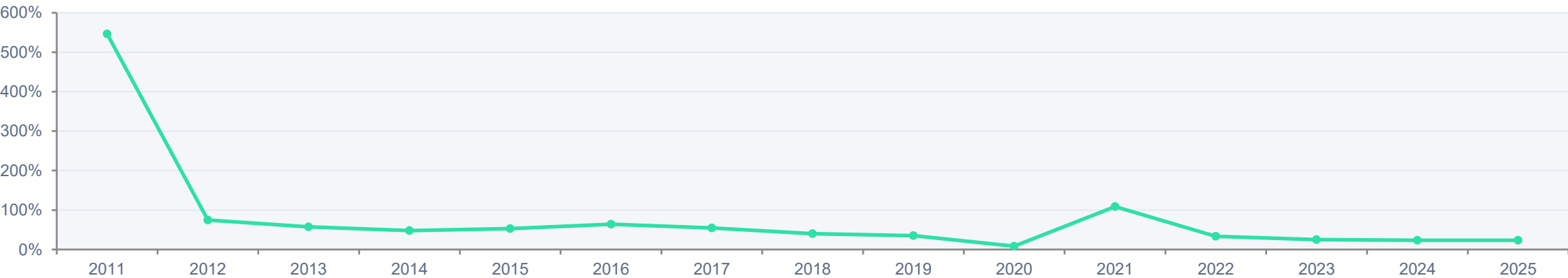
21.00M

EV sales in 2025

2021-2025

Strongest growth period

YEAR-OVER-YEAR EV SALES GROWTH (2011-2025)



KEY MARKET SNAPSHOT

2025 at a Glance

GLOBAL EV SALES

21.00M

GLOBAL EV STOCK

75.00M

EV SALES SHARE

25%

TOTAL CHARGING POINTS

6.97M

KEY MARKET SNAPSHOT — 2025

TOP COUNTRY

China

TOP POWERTRAIN

BEV

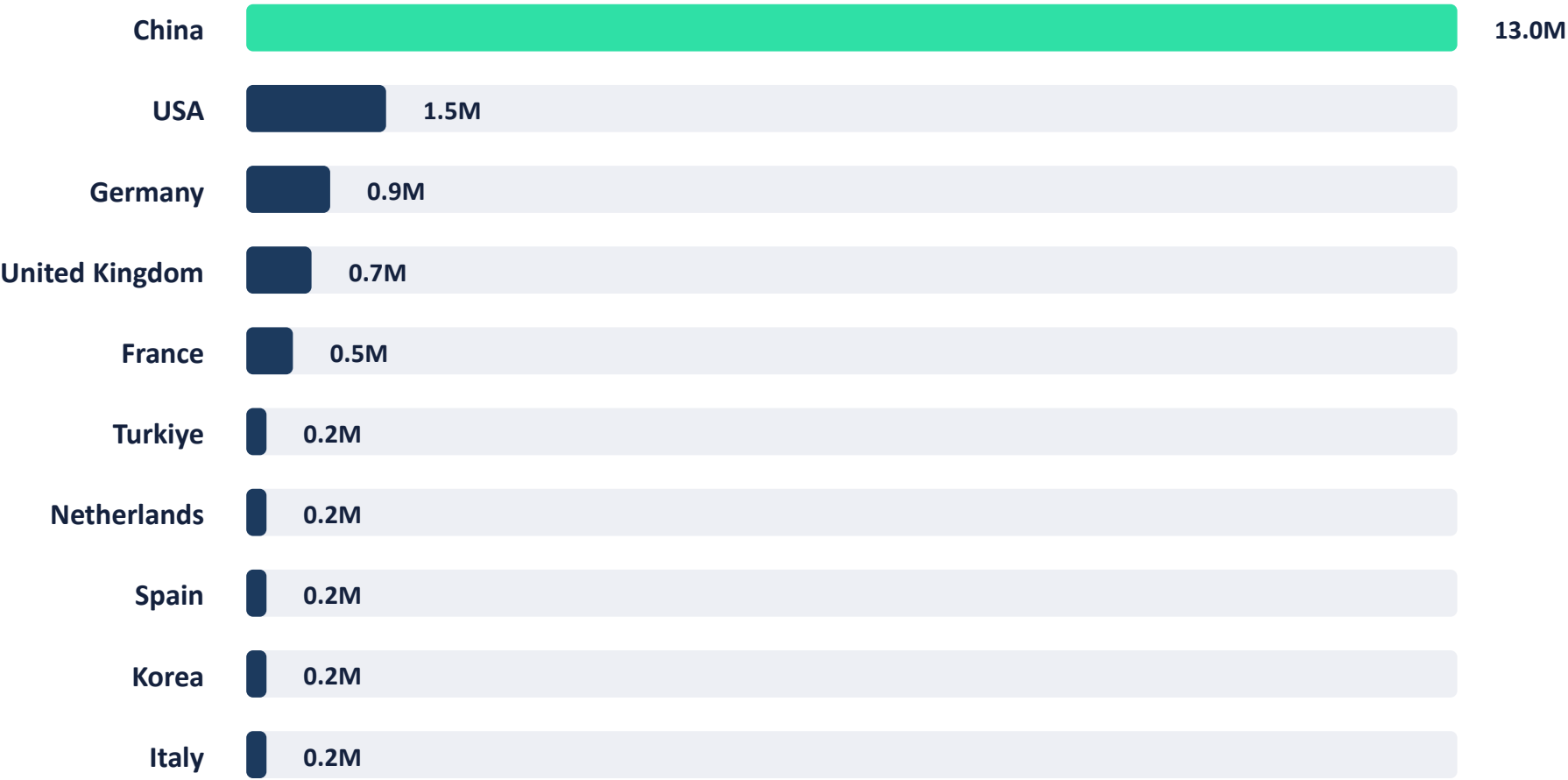
YOY GROWTH %

23.1%

CHARGING STATIONS

6.97M

Which Countries Sell the Most EVs?



Who's Leading, and With What?

CHINA vs USA vs INDIA (HISTORICAL EV SALES)

China



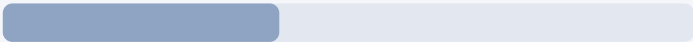
Clear market leader — ~63% of global EV sales

India



Strong recent acceleration

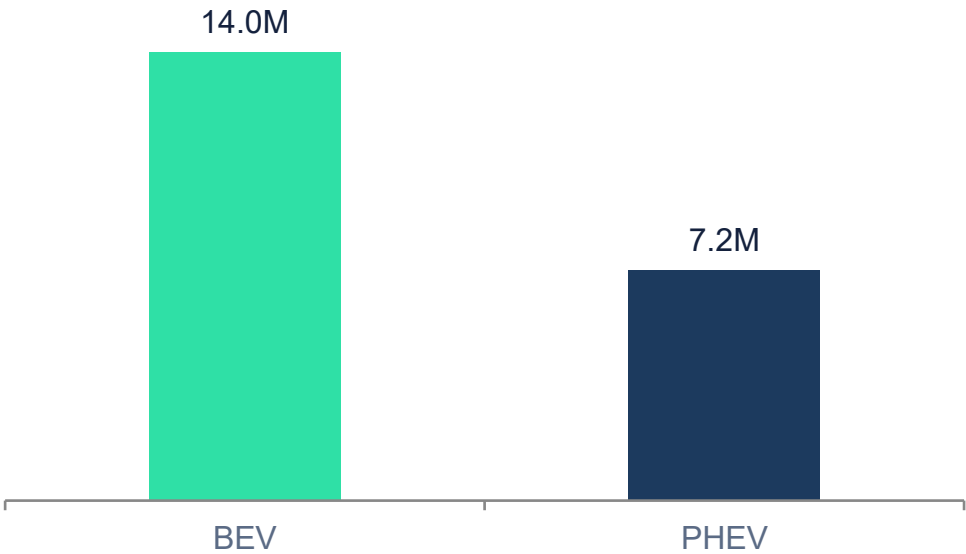
USA



Steady, more moderate growth

China stands out as the dominant EV market, while India shows the strongest recent growth momentum.

BEV vs PHEV SALES DISTRIBUTION — 2025



BEV 14.0M (66.0%) • PHEV 7.2M (34.0%) of total 2025 EV sales.

Where Is EV Demand Concentrated?

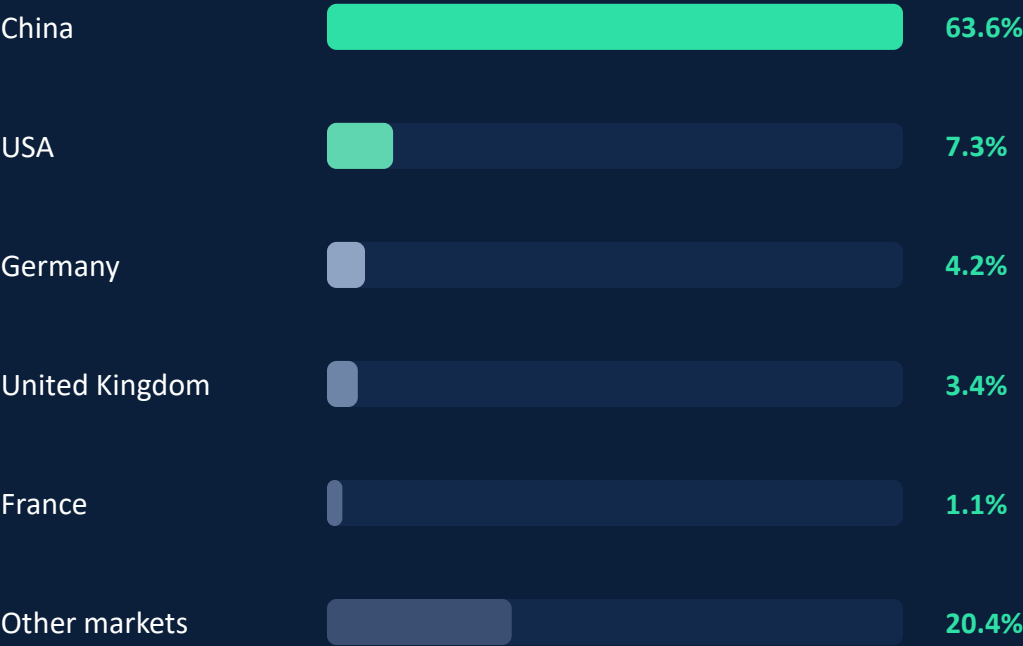


ASIA PACIFIC DOMINATES REGIONALLY

Asia Pacific remains the largest regional EV market in 2025, reinforcing its importance to global EV adoption.

Rising EV sales, growing vehicle stock, and expanding infrastructure indicate electric mobility is becoming increasingly established worldwide.

MARKET CONTRIBUTION BY COUNTRY (2025)



The EV Ecosystem Is Expanding

EV STOCK

39K → **75.00M**

growth in the global EV fleet (2010→2025)

EV MARKET SHARE

0.012% → **25%**

of new passenger-car sales by 2025

FAST CHARGING POINTS

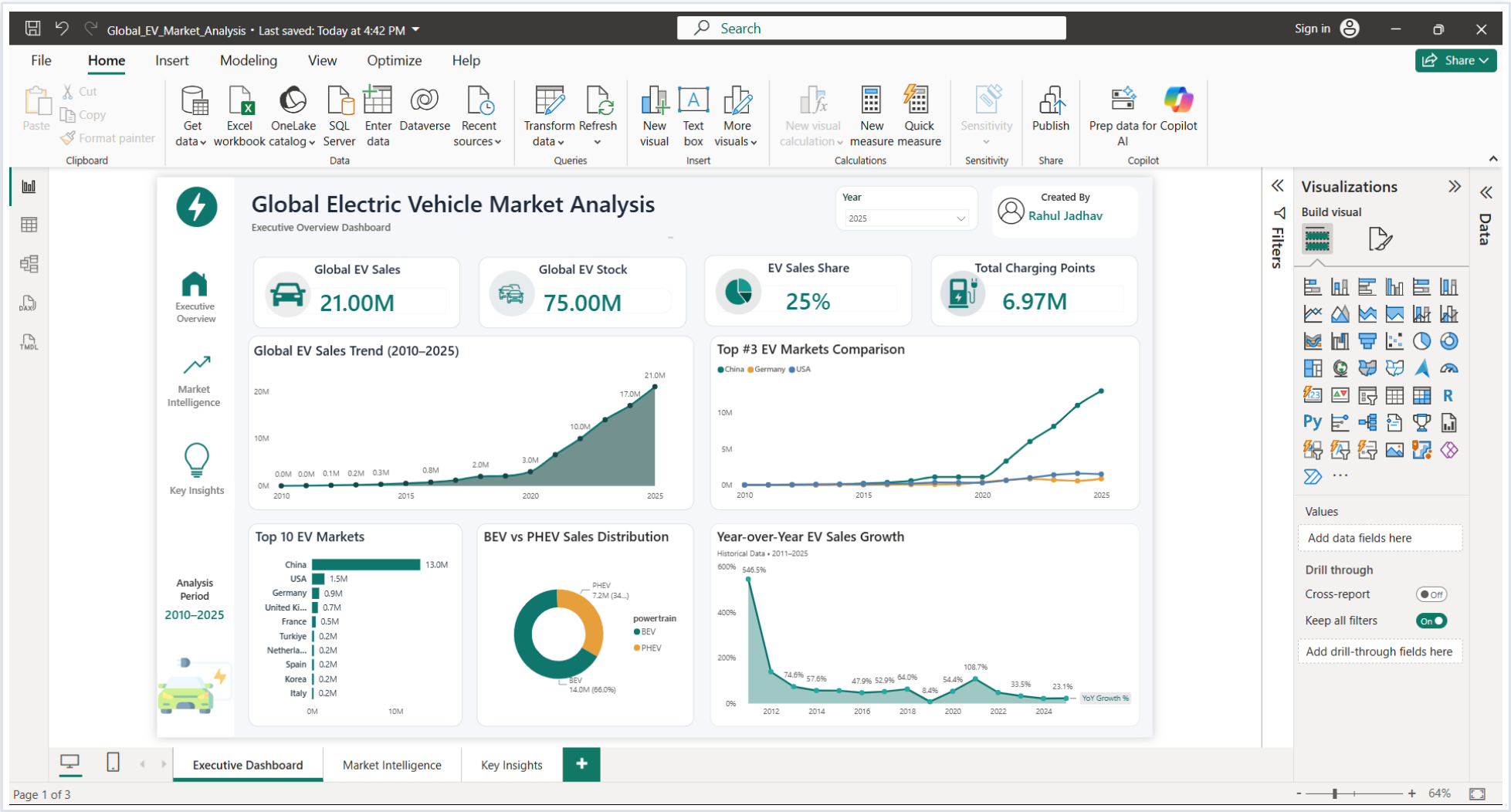
310 → **1.7M**

publicly available, 2010→2025

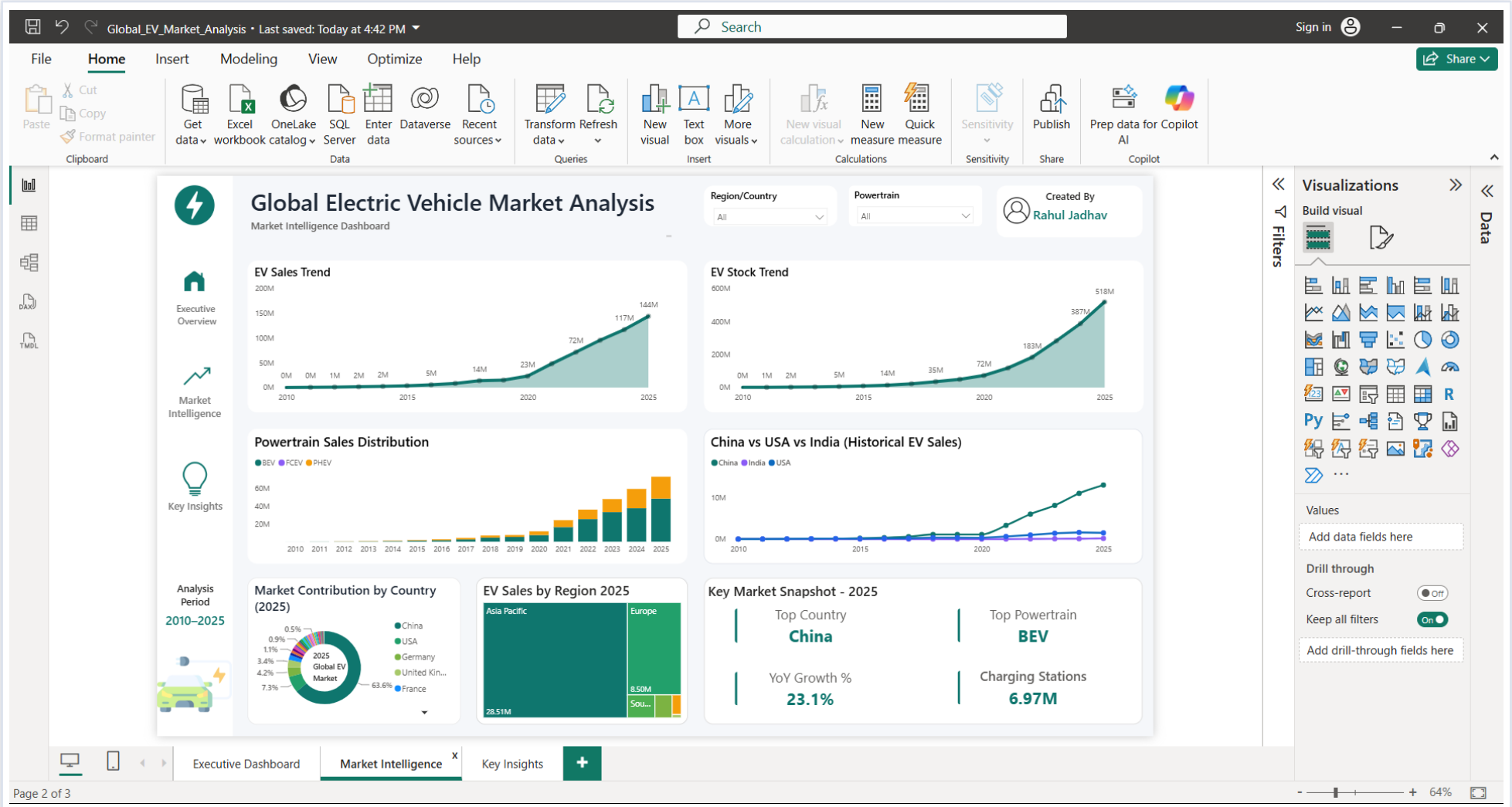


Note: the Power BI dashboard's Total Charging Points KPI (6.97M) counts all charging points in 2025 — a broader measure than the publicly available fast-charging points (1.7M) tracked in the notebook analysis above. The two figures are not interchangeable.

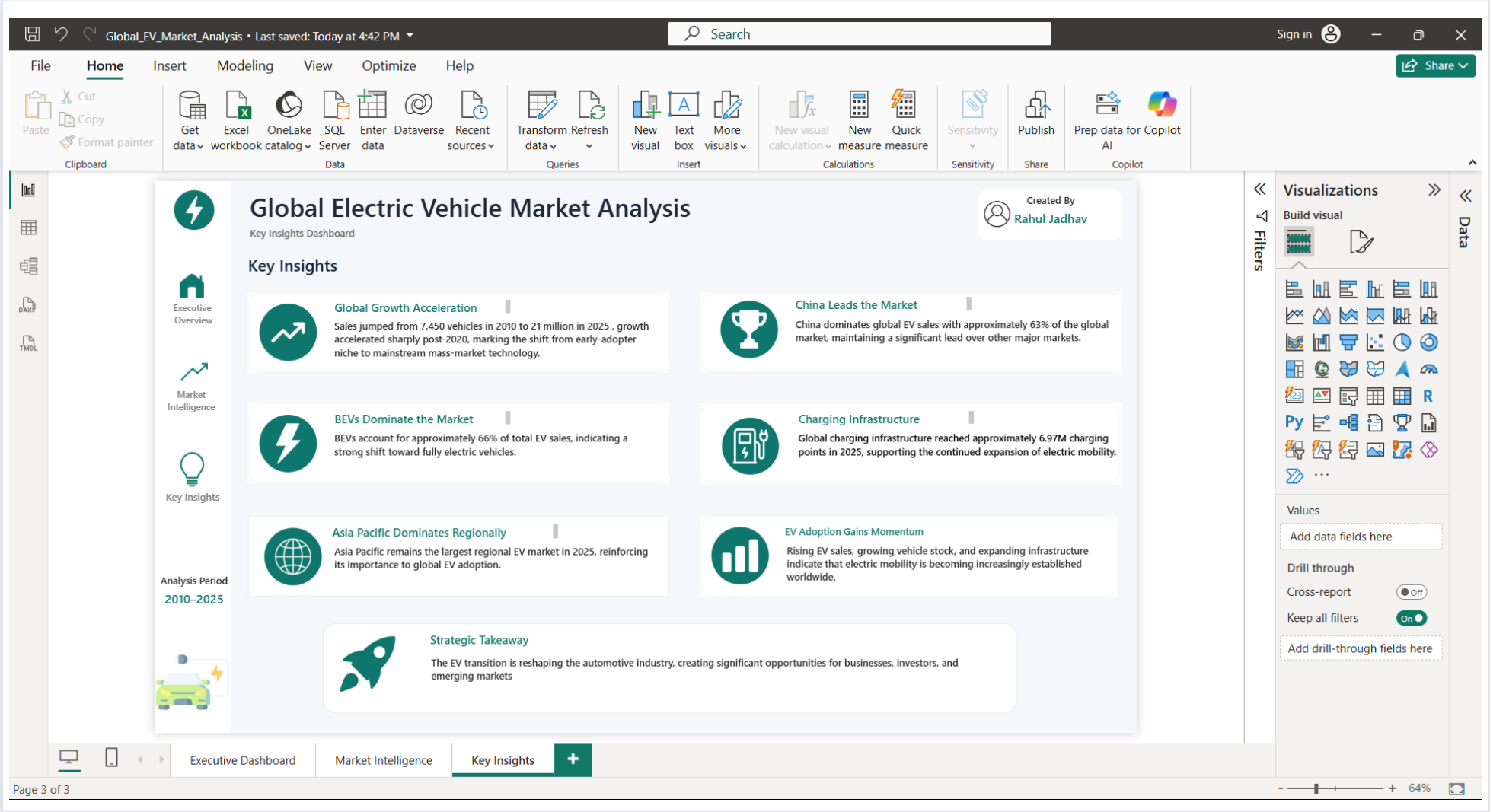
Executive Overview



Market Intelligence



Key Insights



What Does the Analysis Mean?

KEY INSIGHT

Global Growth Acceleration — sales jumped from 7,400 (2010) to 21M (2025), accelerating sharply post-2020

China Leads the Market — ~63% of global EV sales

BEVs Dominate the Market — ~66% of total EV sales

Charging Infrastructure — ~6.97M charging points in 2025

Asia Pacific Dominates Regionally — largest regional EV market in 2025

EV Adoption Gains Momentum — rising sales, stock, and infrastructure together

BUSINESS IMPLICATION

Market has shifted from an early-adopter niche to mainstream mass-market technology

China is the critical market for EV-related businesses

Fully electric vehicles are gaining stronger long-term traction than plug-in hybrids

Infrastructure investment remains essential to sustaining EV adoption

Asia Pacific remains central to global EV strategy

Electric mobility is becoming increasingly established worldwide

STRATEGIC TAKEAWAY

The EV transition is reshaping the automotive industry, creating significant opportunities for businesses, investors, and emerging markets.

THANK YOU

Let's Connect



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GITHUB

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